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RDS Document
Requirement & Design Specification

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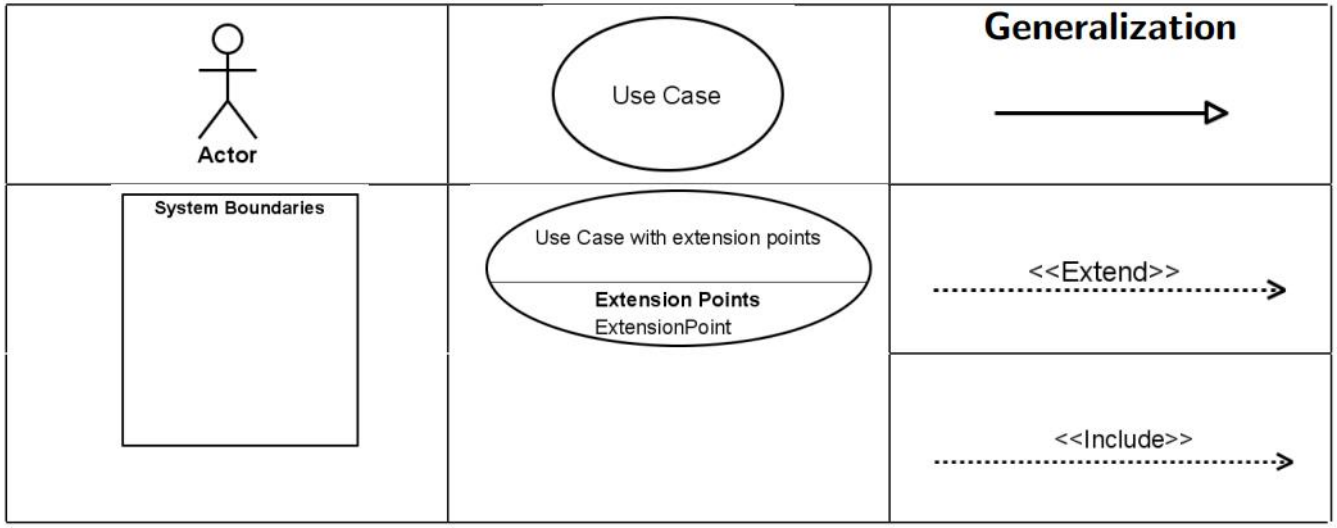
Nội dung bài học

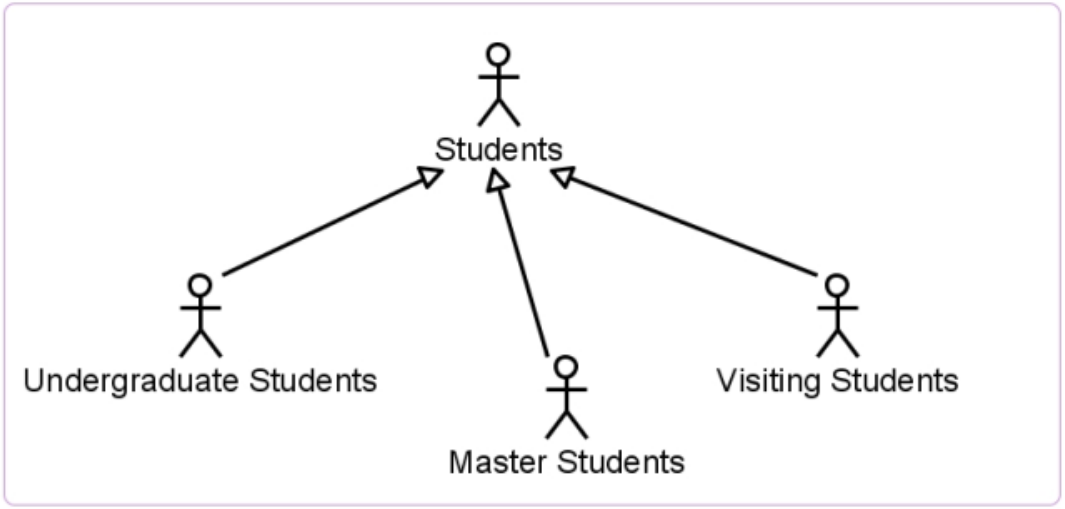
- Use cases diagram
- Screens Flow
- System High Level Design
- Requirement Specifications

Use cases diagram

- Actors
  - An actor is a person (or sometimes another software system or a hardware device) that interacts with the system to perform a use case. Following are some questions you might ask to help user representatives identify actors
    - Who (or what) is notified when something occurs within the system?
    - Who (or what) provides information or services to the system?
    - Who (or what) helps the system respond to and complete a task?
- Use Cases
  - A use case (UC) describes a sequence of interactions between a system and an external actor that results in the actor being able to achieve some outcome of value. The names of use cases are always written in the form of a verb followed by an object. Select strong, descriptive names to make it evident from the name that the use case will deliver something valuable for some user

Use cases diagram (1)





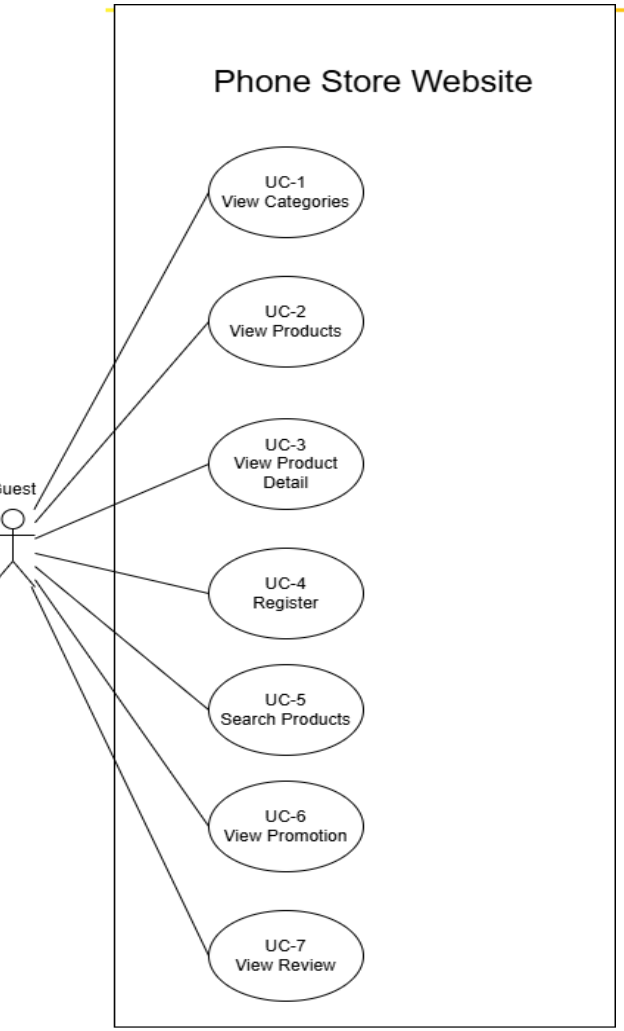
Use cases diagram (2) Example

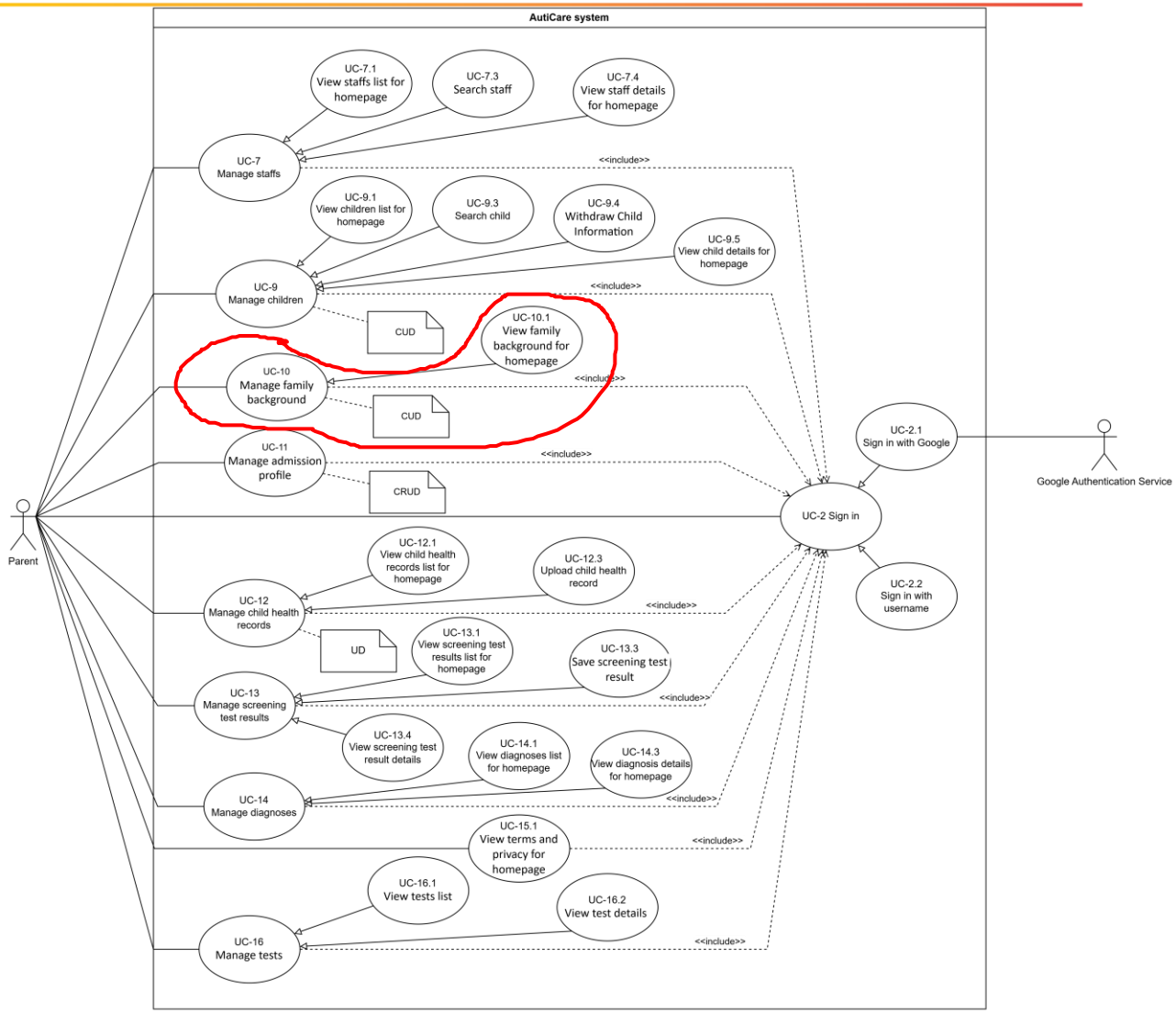
- SMARTLAB - EXPERIMENTAL DATA MANAGEMENT SYSTEM IN AGRICULTURE

#	Actor	Description
1	Guest	Guests are entirely new users of the system who have never registered an account. Guests can view information about the system and register for an account. Additionally, guests can utilize the AI-powered disease detection functionality provided by the system and view information on the identification and treatment of these diseases.
2	User	Users are new individuals commencing their journey with the system after registered and email-verified successful. They gain access to system-wide functionalities related to profile management in addition to their existing capabilities. Furthermore, Users can upgrade their role to Lab Owner through payment.
3	Client	Clients within the system are essentially users, but they are designated as customers of specific experiments when added by Lab Owners. As clients of these experiments, they gain access to information regarding batches, samples, tests, and reports within the respective experiments. Additionally, to facilitate communication between clients and other members involved in the experiment, clients can engage in direct commenting on the samples of interest, enabling seamless exchange of information and feedback.
4	Analyst	Analysts within the system are users who are added to experiments by Lab Owners with the role of Analyst. Once users have confirmed their email addresses, they can participate in managing samples and tests within the assigned experiment. For analysts who have previously upgraded their system package but have expired, they are limited to viewing basic information about the laboratory and experiments.
5	Standard User	As a standard user, individuals in the system serve as basic users with the flexibility to assume multiple roles based on the experiment they are managing.
6	Laboratory Owner	As Lab Owners, users who have upgraded their system package and are within the active period gain extensive privileges within the system. They have full authority to manage all information related to their laboratory and experiments. Additionally, Lab Owners can monitor all activities within the experiment and can add or remove members from the experiment as needed. Furthermore, Lab Owners can generate reports to review the samples and tests present within the experiment.
7	Admin	Admin is the person responsible for managing labotories, account and subscriptions of the system.

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Use cases diagram (3) Example





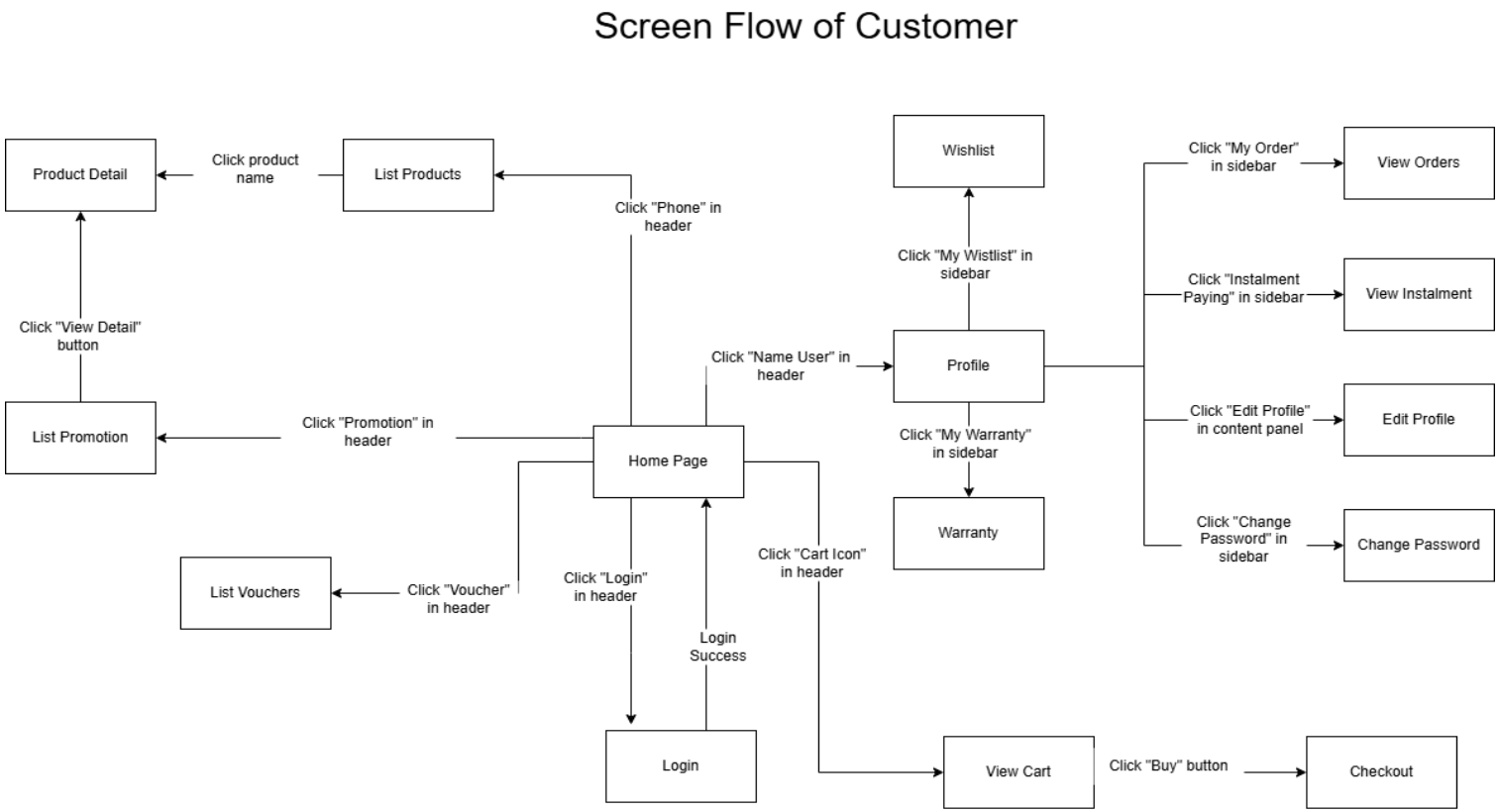
Use cases diagram (4) Example

ID	Feature	Use Case	Use Case Description
UC-10.1	Manage Family background	View family background for homepage	….
UC-10.2	Manage Family background	Add family background for homepage	…
UC-10.3	Manage Family background	Update family background	…
UC-10.4	Manage Family background	Delete family background	…

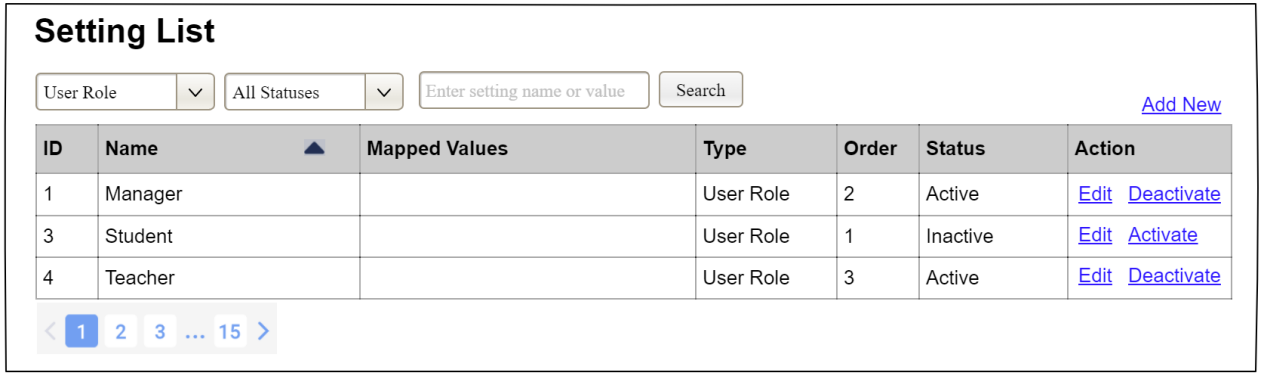
Screens Flow

- This part shows the system screens and the relationship among screens. You can draw the Screens Flow for the system in the form of diagram as below. Please note that beside the normal flat screen, we might have the oval notation for pop-up screen (Import Order) or a screen with multiple information tab (Order Details), etc. You may also use text or background format for different visuality purpose.

Screens Flow (1) Example



Screens Flow (2) Example



2.2 Screen Descriptions

#	Feature	Screen	Description
1	View list role	Setting List	<<Screen Brief description>>
2	Add new	Setting List	<<Screen Brief description>>
3	Edit role	Setting List	<<Screen Brief description>>
4	Active/Deactive	Setting List	<<Screen Brief description>>

Screens Flow (3)

- 2.3 Screen Authorization

Screen/Function	Role-Name1	Role-Name2	Role-Name3	…
<<Screen Name1>>	X		X	X
<<Screen Activity>>			X	X
Setting List	X		X	
View list role	X			
Add new			X	
Edit role			X	
Active/Deactive			X	X
…				

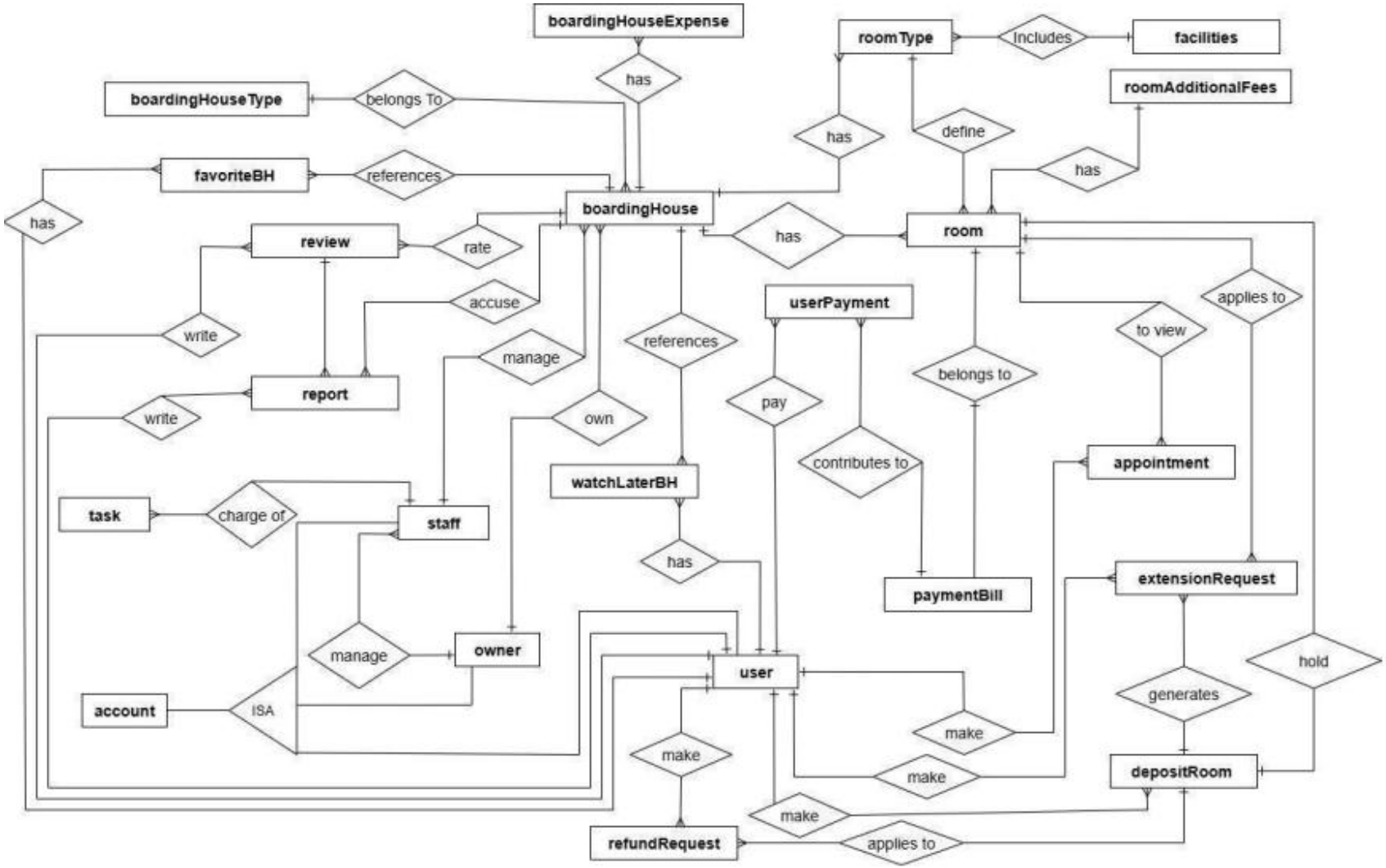
Screens Flow (3)
2.4 Non-UI Functions

- Provide the descriptions for the functions which have no UI (or not screens), i.e batch/cron job, service, API, etc.
- Example

#	Feature	System Function	Description
1	Email Verification	Verified email	The system authenticates the user's email to grant permission to log into the system.
2	Confirm to Join Experiments	Confirm to Join Experiments	The system will send a confirmation email that has been confirmed in the experiment, the customer can check in and join the experiment
3	Payment	Upgrade service in Vn Pay (Externals)	Users can pay for an upgrade package to be able to use advanced functions of the system
4	Upgrade to Laboratory Owner	Auto opening all experiments of upgraded account's laboratory	In the system, if a Lab Owner's subscription expires, the experiments within their lab will be closed. However, if they upgrade to a different package and the new package is still active, the system will automatically reopen all experiments within their lab. This allows them to continue managing and conducting research activities without interruption.

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System High Level Design
3.1. ERD



System High Level Design
3.2. Database Design

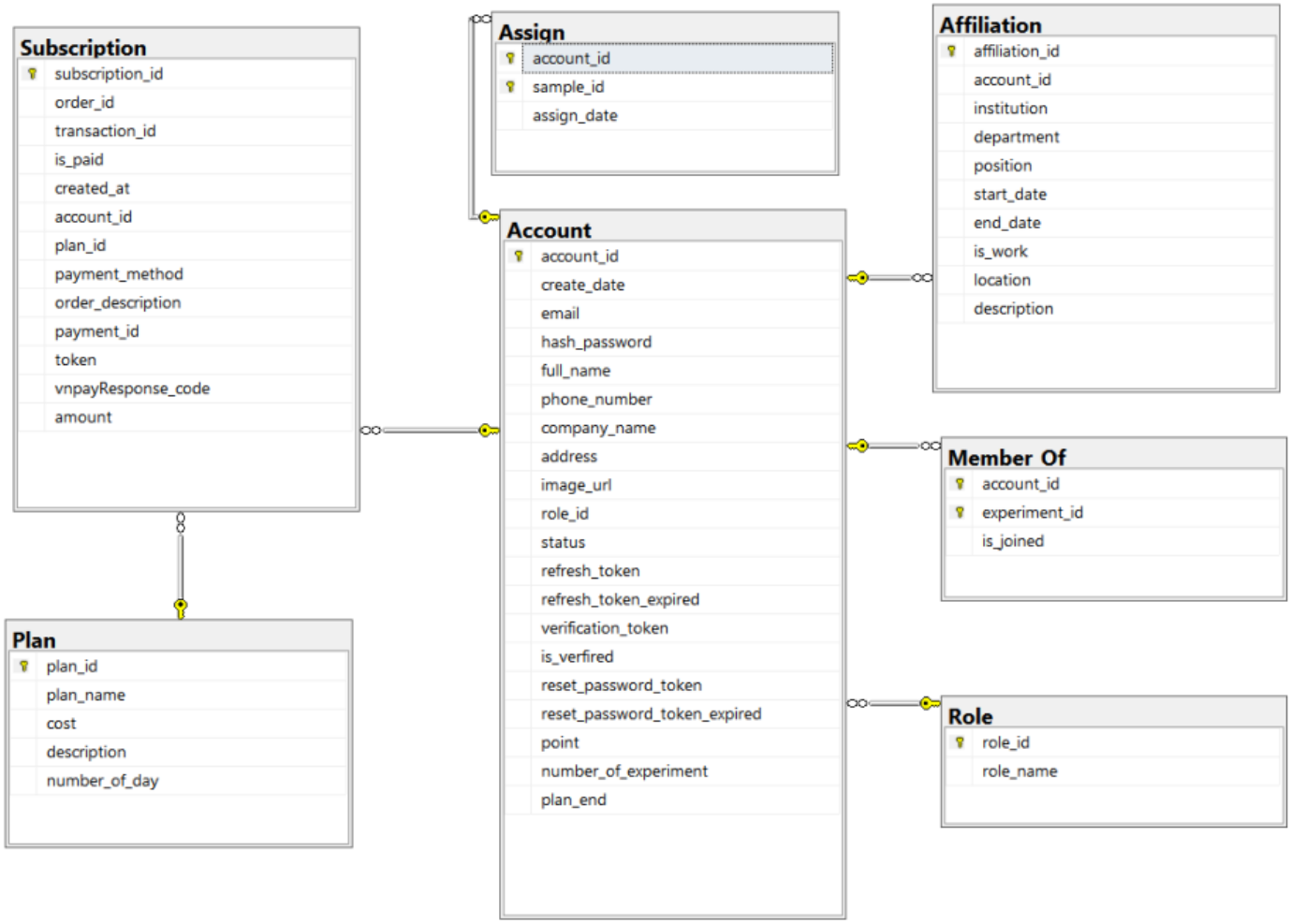


Figure 43. Account service database

System High Level Design
3.2. Database Design

- *Table Descriptions*

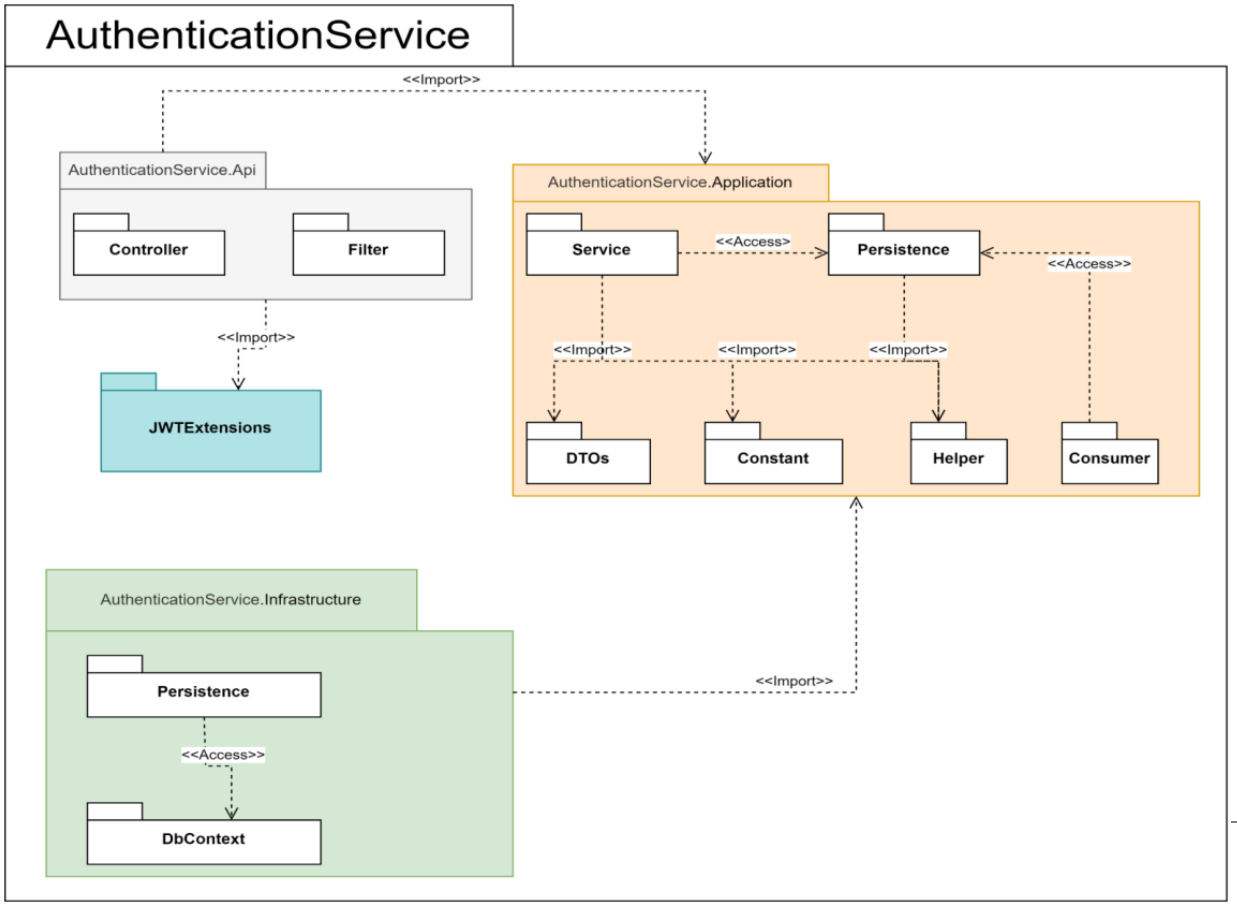
No	Table	Description
01	*<Table name>*	*<Description of the table>* *- Primary keys: <<list of primary key fields>>* *- Foreign keys: <<list of foreign key fields>>*
02	*<Table name2>*	---

Table Descriptions

No	Table	Description
01	*Account*	*- Primary keys: account_id: uniqueidentifier* *- Foreign keys: role_id: int* *- Attributes:* • *create_date: datetime* • *email: nvarchar(MAX)* • *hash_password: nvarchar(MAX)* • *full_name: nvarchar(MAX)* • *phone_number: nvarchar(MAX)* • *company_name: nvarchar(MAX)* • *address: nvarchar(MAX)* • *image_url: nvarchar(MAX)* • *status: bit* • *refresh_token: nvarchar(MAX)* • *refresh_token_expired: datetime*

System High Level Design
3.3 Code Packages

- Provide the package diagram for each sub-system. The content of this section including the overall package diagram, the explanation, package and class naming conventions in each package. Please see the sample & description table format below



System High Level Design
3.3 Code Packages

- Package descriptions

Package descriptions

No	Package	Description
01	*Member_authority*	*<Description of the package>*
02	*registration*	*<Description of the package>*
03	*...*	

Requirement Specifications

- Provide the functional description for the use cases using the template/guides below

Functional Description Template

UC ID and Name:			
Created By:		Date Created:	
Primary Actor:		Secondary Actors:	
Trigger:			
Description:			
Preconditions:			
Postconditions:			
Normal Flow:			
Alternative Flows:			
Exceptions:			
Priority:	High (Medium, Low), Must Have (Should Have, Could Have),..		
Frequency of Use:			
Business Rules:	BR01		
Other Information:			
Assumptions:			

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Requirement Specifications

UC ID and Name:	**UC-2_Login System**		
Created By:	MinhNNT	Date Created:	16/Jun/2023
Primary Actor:	Customer	Secondary Actors:	None
Trigger:	User clicks Login button from the page header, or User accesses an authenticated feature (from a link or type the page URL directly into the address bar)		
Description:	As a user, I want to be able to log into the system so that I can use the system's authenticated features and access my personalized account.		
Preconditions:	User account has been created & authorized		
Postconditions:	• User logs in the system successfully • The system tracked successful login into the Activity Log		
Normal Flow	**2.0 Login System** 1. User accesses the User Login screen 2. User types in the login details or choo other login options (see 2.1 and 2.2) 3. User clicks the Login button 4. System validates the login details (see 2.0.E1) 5. System allows user to access 6. System tracks user's success login to the Activity Log 7. System accesses the Home Page (or the previous calling page if any)		
Alternative Flows:	***2.1 Google Login*** 1. User chooses to login system using Google account 2. System redirects the user to the Google's Login screen 3. User types in the Google account details and chooses to login 4. Google validates user's login information successfully and redirect him/her back to the system 5. Return to step 5 of normal flow. ***2.2 Facebook Login*** 1. User chooses to login system using Facebook account 2. System redirects the user to the Facebook's Login screen 3. User types in the Facebook account details and chooses to login 4. Facebook validates user's login information successfully and redirect him/her back to the system 5. Return to step 5 of normal flow.		
Exceptions:	***2.0.E1 System can't authenticate the user*** 1. The Error Message screen is shown to the user 2. User cancels the logging in => *UC stops, change to UC-1_View Home Page* 3. User clicks "Forgot Password?" link => *change to UC-3_Reset Password* 4. User clicks "Register" link => *change to UC-4_Register User Account*		
Priority:	Must Have		
Frequency of Use:			
Business Rules:	FR1, FR2, FR3		

b. Business Rules

ID	Business Rule	Business Rule Description
BR1	Password Encoding	User's password must be encoded with MD5 hashing
BR2	Invalid Logging In	User can't be authenticated to login the system if below cases • His/her logging-in details are incorrect • His/her account has not been verified • His/her account has been locked or blocked
BR3	Account Locking	If user inputs wrong logging-in details 6 times continuously, his/her account would be locked in 30 minutes

Requirement Specifications

3.2 UC-6_Register for Payroll Deduction

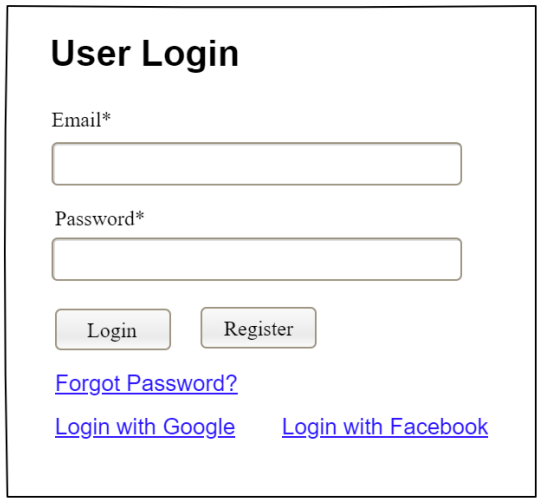
a. Functional Description

ID and Name:	**UC-6 Register for Payroll Deduction**		
Created By:	Nancy Anderson	Date Created:	9/15/13
Primary Actor:	Patron	Secondary Actors:	Payroll System
Description:	Cafeteria patrons who use the COS and have meals delivered must be registered for payroll deduction. For noncash purchases made through the COS, the cafeteria will issue a payment request to the Payroll System, which will deduct the meal costs from the next scheduled employee payday direct deposit.		
Trigger:	Patron requests to register for payroll deduction, or Patron says yes when COS asks if he wants to register		
Preconditions:	PRE-1. Patron is logged into COS.		
Postconditions:	POST-2. Patron is registered for payroll deduction.		
Normal Flow:	**6.0 Register for Payroll Deduction** 1. COS asks Payroll System if Patron is eligible to register for payroll deduction. 2. Payroll System confirms that Patron is eligible to register for payroll deduction. 3. COS asks Patron to confirm his desire to register for payroll deduction. 4. If so, COS asks Payroll System to establish payroll deduction for Patron. 5. Payroll System confirms that payroll deduction is established. 6. COS informs Patron that payroll deduction is established.		
Alternative Flows:	None		
Exceptions:	6.0.E1 Patron is not eligible for payroll deduction 6.0.E2 Patron is already enrolled for payroll deduction		
Priority:	High		
Business Rules:	BR-86 and BR-88 govern an employee's eligibility to enroll for payroll deduction.		
Other Information:	Expect high frequency of executing this use case within first 2 weeks after system is released.		

Design Specifications

- Provide brief description of the screen/function + related UC here and other details as in the sub-sections
- UI Design
  - This is to describe the UI layout (Mockup prototype) & descriptions for screen fields/components
- Database Access
  - Provide the design description for the screen/function to access the database here: what table the screen/function would access, which transactions does it make (C-Create, R-Read, U-Update, or D-Delete), and how/purpose of the access (by providing Description and SQL commands)
- SQL Commands
  - Provide the detailed SQL (select, insert, update...) which are used in implementing the screen/function

Design Specifications



Field Name	Field Type	Description
Email*	Text Box	This is for user to input valid email address for logging in
Password*	Password Box	This is for user to input password for logging in
Login	Button	User clicks to authenticate him/herself into the system with provided email & password
Register	Button	User clicks to redirect to the User Register page for registering new user account to access the system
Forgot Password?	Hyperlink	User clicks to redirect to the Password Reset page for resetting his/her forgot password
Login with Google	Hyperlink	Allow user to login with his/her Google account
Login with Facebook	Hyperlink	Allow user to login with his/her Facebook account

Database Access

Table	CRUD	Description
User	R	Verify UserName & Password information
Setting, User	R	Specify the authorizations of the logged-in user

SQL Commands:

1/ Verify UserName & Password information

```
SELECT user_id, full_name, email, image_url, status
FROM user WHERE user_name = ? AND password = ?
```

2/ Specify the authorizations of the logged-in user

```
SELECT mapped_values FROM setting WHERE setting_id = ?
SELECT setting_name, mapped_values FROM setting WHERE setting_id IN (?)
```

Nhiệm vụ

- Chọn nhóm của mình
- Tìm hiểu về vấn đề sẽ làm, trong đó:
  - Xác định được hệ thống sẽ xây dựng
  - Tìm hệ thống liên quan để xác định: actor, usecase,…
- Xây dựng diagram liên quan trong template 2 – RDS để:
  - Nắm được chức năng sẽ làm.
  - Có diagram để GV sửa + hiểu và điều chỉnh

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